Career Compass

A Skill-Based Job Recommendation Platform

Team Workforce Insights - SIADS 699 Capstone

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The Problem

Job postings describe the same skills and qualifications using wildly inconsistent terminology. This makes it hard for job seekers to compare roles, recognize a strong fit, or identify the specific skills standing between them and a role they want.

Our Approach

Career Compass extracts real skills directly from unstructured job-description text, then recommends postings based on a user's skill profile and surfaces the specific skills they're missing relative to similar roles - using only what's in the posting itself, since no resume database or user click history exists for this dataset.

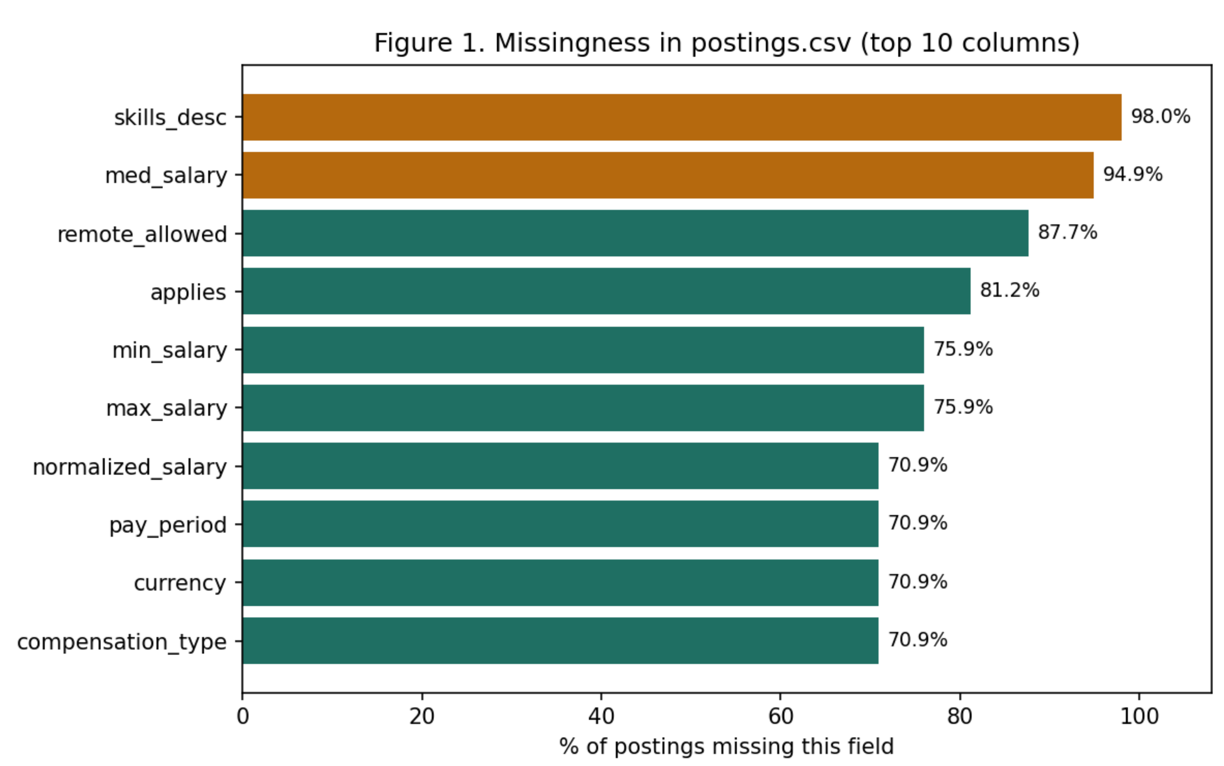
The Dataset

123,849
LinkedIn job postings analyzed

133
skills recognized across 10 categories

88.6%
of postings matched at least one skill

24,473
companies in the supporting data



Several fields we hoped to use directly are mostly empty - this is what forced the skill-extraction approach below.

Methodology

1. Explore
Two EDA passes surfaced severe missingness (skills_desc 98% empty; salary 71-95% missing) and confirmed structured skill fields cover only 35 broad categories, not real skills.

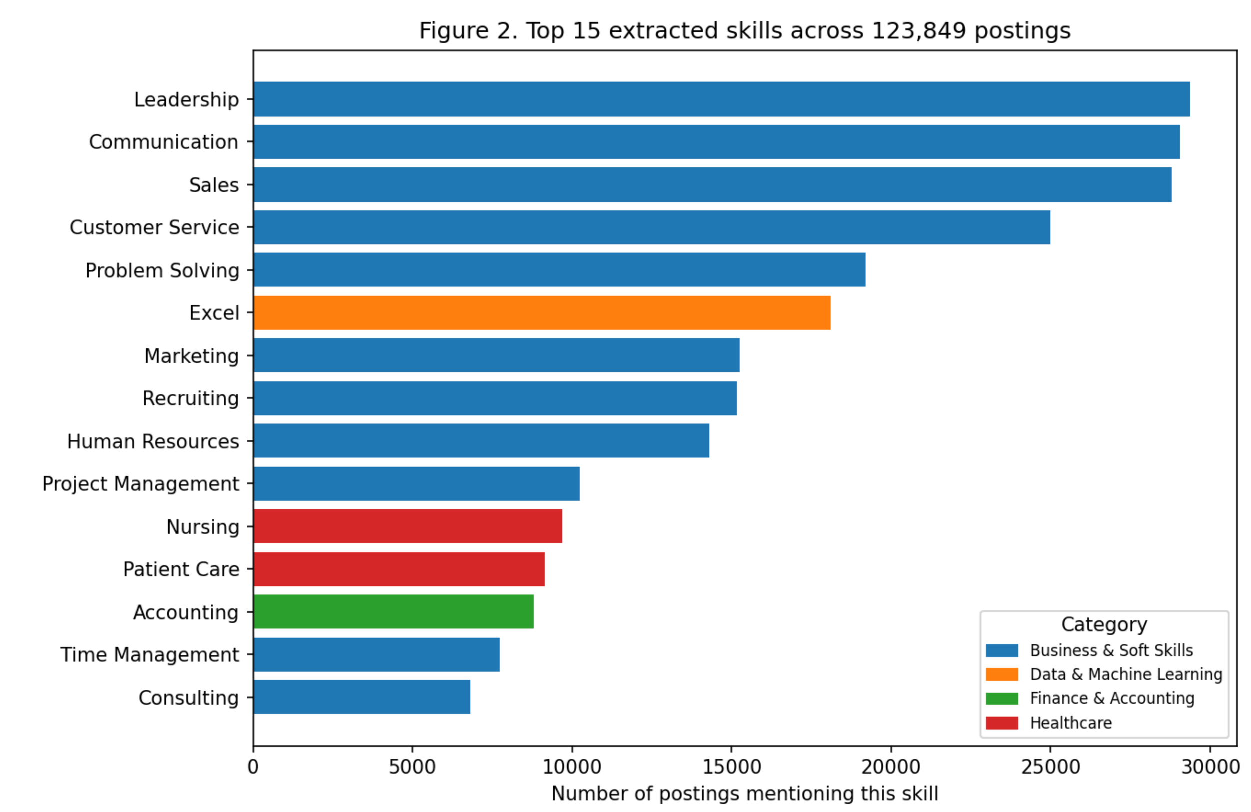
2. Clean
Resolved every open question with evidence: derived is_remote (no explicit false value exists in the raw data), flagged 1.3% of salary rows as outliers, fixed placeholder nulls in company data.

3. Extract
Built a 133-skill gazetteer matched with flashtext, a trie-based algorithm - after an initial regex approach proved too slow at 123,849 documents (would have taken ~1 hour; flashtext: 95 seconds).

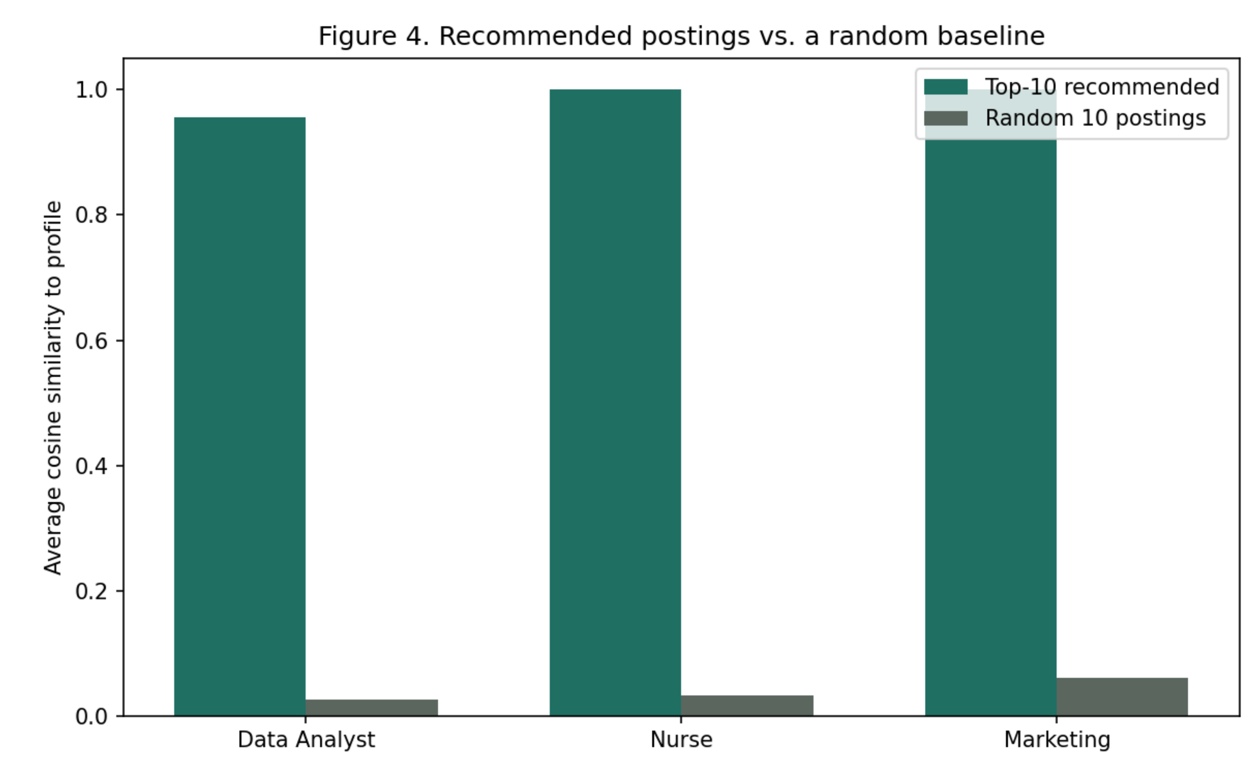
4. Recommend
IDF-weighted cosine similarity over skill vectors, so rare specific skills (AutoCAD, HIPAA) count for more than ubiquitous soft skills (Leadership, Communication) that would otherwise dominate the match.

5. Deliver
A Streamlit dashboard where a user selects skills and receives ranked postings plus a skill-gap list, in real time.

What the Data Shows



Top 15 extracted skills. Broad soft skills dominate since the dataset spans every industry, not just tech - exactly why IDF weighting matters for recommendation quality.



Top-10 recommendations score 16-37x higher in average similarity than a random baseline across three test profiles - confirming the ranking does substantive, non-arbitrary work.

Why It Matters

The clearest beneficiaries are job seekers navigating a market outside their immediate network, plus career counselors who could use skill-gap patterns to inform advising. Every recommendation is explainable in terms of specific matched skills - not a black-box score - so a user can see exactly why a posting was suggested.

An Honest Limitation

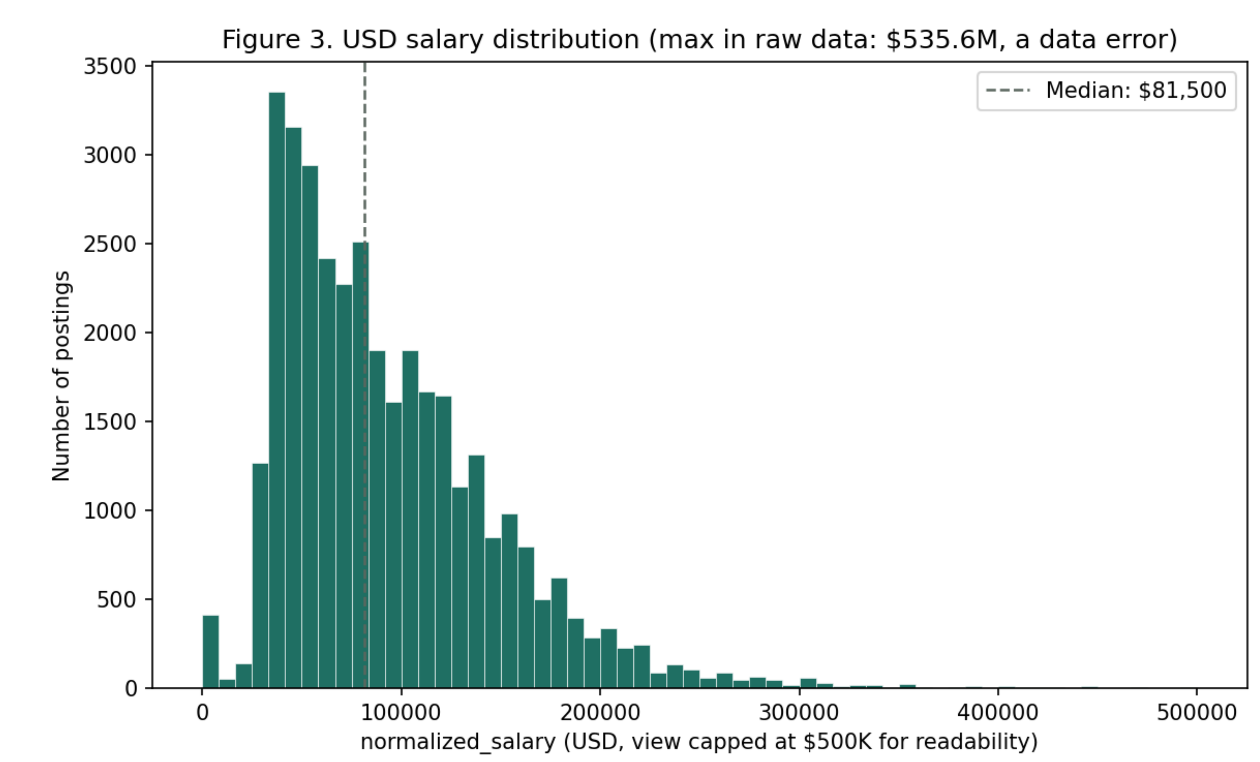
Job postings can reflect the hiring biases of the employers who wrote them (see, e.g., Amazon's discontinued hiring algorithm, which learned to downrank resumes associated with women). We have not audited this dataset for that pattern. We mitigate what we can control: full transparency in why a posting is recommended, and skill gaps framed as informational context, not a qualification verdict.

What's Next

Expand the skill gazetteer with an established taxonomy (O*NET/ESCO) or a trained NER model; add result-diversity controls; evaluate with real users rather than only a random baseline.

Code, notebooks, dashboard & full report:

github.com/lilmeem/
team-workforce-insights



A reminder to check your data: raw salary max was $535.6M against a median of $81,500 - a clear entry error we flagged rather than trusted.